

Survivor Bias in Learning-Curve Surrogates: Successive Halving Selects on Noise

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Abstract

Learning-curve surrogates for hyperparameter optimization — freeze-thaw GPs, power-law extrapolation, in-context learning-curve networks — are fit and deployed on curves that *survived* earlier rungs of successive halving (SH), Hyperband, or ASHA, yet no method in this line models the censoring mechanism. We state the honest crux up front: selection acts on *observed* prefixes, so if prefixes were noiseless the censoring would be ignorable (missing at random). The bias enters because prefixes are noisy observations of latent curve parameters: rung survival selects on noise, so survivors’ continuations systematically disappoint a naive per-curve extrapolation — regression-to-the-mean survivor bias with a Tobit/Heckman structure on the latent parameters. In a controlled power-law family the effect is precisely dose-responsive: survivor selection bias at the first rung grows from 0.0012 (accuracy units) at noise $\sigma = 0.005$ to 0.0539 at $\sigma = 0.04$ and grows with selection pressure η ; at the *median* noise we calibrate from real LCBench curves ($\sigma \approx 0.0004$) it is *indistinguishable from zero* — an informative null about SH robustness that we report rather than bury, with the caveat that noisier regimes (small validation sets; PD1’s ImageNet task at $\sigma \approx 0.02$) sit well inside the biased zone. A Heckman-type joint selection+outcome model — a hierarchical prior over latent curve parameters with rung-survival probit likelihoods — attacks the survivor bias but exposes the same identification fragility econometrics warns of: with a single SH bracket it is weakly identified (it prefers the biased solution at realistic survivor counts), and pooling three Hyperband-style brackets — rung-assignment randomness as an *exclusion restriction* — cuts the survivor-only asymptote bias from +0.084 to −0.036 accuracy units (naive 0.864, corrected 0.744, truth 0.780) but now *overshoots*, more so at higher noise. The robust practical estimator is the boring one: refit the prior on *all* observed prefixes, including those of killed configs (0.784, near-unbiased) — exactly the data a replaying tool already holds, reinforcing the crux that the fix is to use the prefixes one has, not to model survivor selection. The same posterior machinery cuts survivor-extrapolation bias at the first rung from 0.0192 (naive least squares) to 0.0057; a per-curve Tobit correction without the population prior manages only 0.0144, and an Arellano–Bond-flavoured lag-instrument extrapolator is worse than the naive fit and is dropped. On LCBench and PD1 replays the story is honestly mixed and matches the calibrated-noise prediction: plain last-value SH is already near-optimal at decision level; naive power-law extrapolation *hurts* promotions and suffers a winner’s curse when picking from extrapolations (top-decile prediction bias 0.183 on LCBench); the corrected surrogate slashes that tail bias ($\sim 7\times$, to 0.026) and ties last-value SH on LCBench decisions, but on PD1 it improves prediction while *worsening* decisions, and a one-parameter variance-penalty hack captures most of the benefit for far less machinery — a prediction–decision gap we quantify rather than paper over.

1 Introduction

Multi-fidelity hyperparameter optimization kills most of its training runs early: successive halving [9], Hyperband [13], and ASHA [12] advance only the top $1/\eta$ of configurations at each rung. An active literature fits *learning-curve surrogates* to the resulting prefixes — freeze-thaw GPs [18], parametric extrapolation [3, 11], deep-kernel multi-fidelity models [21], power-law scaling fits [10], and in-context curve extrapolators [1, 17]. None of these models the sampler’s censoring mechanism, although every deployed curve the surrogate ever sees was selected by it. This remains true in the newest work: a 2025 method that predicts learning curves

with latent-Kronecker GPs to *reduce* premature pruning [14] still trains its predictor on survivor curves as if they were an unbiased sample, and a fresh large-scale learning-curve database documents that curves are more ill-behaved than assumed [15] without connecting the ill-behavedness to selection. The gap our pre-registered kill-test found (§8) holds.

The honest crux. Stated immediately, because it disarms the obvious objection: SH selects on the *observed* prefix, which the surrogate also observes, so if prefixes were noiseless functions of the latent curve the censoring would be ignorable — missing at random, nothing to correct. The bias enters through observation noise. A config survives the rung at epoch t_k if its *noisy* value there clears the bar; survival therefore selects on the noise realization ($\mathbb{E}[\epsilon_{t_k} \mid \text{survived}] = 0.0056$ at $\sigma = 0.02$, $\eta = 3$ in our controlled family), the fitted latent parameters of survivors are noise-inflated, and their continuations disappoint the extrapolation — Galton’s regression to the mean [5], in the form of a Tobit/Heckman selection structure [6, 19] on the *latent* curve parameters, with Tweedie/empirical-Bayes shrinkage [4] as the natural repair. This is a precise, testable claim, not a vibe, and it makes two falsifiable predictions: the bias must grow with noise and with selection pressure, and it must vanish as noise $\rightarrow 0$. Both hold (§3).

Identification, up front (shared with the companion paper 8). Recovering the *unselected* population of curves from survivor-only data is a Heckman problem, and without an exclusion restriction it rides on functional form. HPO has a natural instrument: *rung-assignment randomness*. Hyperband runs identical populations under brackets with different rung schedules; bracket assignment affects survival but not the latent curve. We show empirically that the survival-corrected estimator is weakly identified from one bracket and roughly halves the bias from three — though it then overshoots, so the estimator we actually recommend is the MAR one that uses killed configs’ prefixes (§4).

Contributions.

1. **Mechanism, dose-response, and an informative null (§3):** in a controlled power-law family under SH, survivor bias of naive extrapolation at the first rung grows from 0.0012 to 0.0539 accuracy units as σ goes $0.005 \rightarrow 0.04$ (and with η : $0.0165 \rightarrow 0.0279$); at LCBench’s *median* calibrated noise the bias is $\approx$ zero. SH is robust exactly where its inputs are clean — and several real regimes are not (PD1 ImageNet: $\sigma \approx 0.02$).
2. **A corrected surrogate with an honest identification story (§4):** a hierarchical Gaussian prior over pow3 parameters in a Laplace/sufficient-statistic formulation with rung-survival probit terms. Survivor-only + single bracket: weakly identified (we show the likelihood prefers the biased solution). Survivor-only + three brackets: bias roughly halved but overshooting (0.744 vs truth 0.780, survivor-naive 0.864). The estimator we actually recommend includes killed-curve prefixes (the data a replaying tool has anyway) and is near-unbiased (MAR reference 0.784) — the theory’s real payoff is not a survivor-only correction but the diagnosis of *which* data to fit on.
3. **Decision-level replays on LCBench and PD1 (§5):** naive power-law extrapolation *hurts* SH (regret 0.0057 vs 0.0020 for last-value on LCBench pools); the corrected surrogate pulls it back to last-value parity there; the winner’s curse of picking from extrapolations is large (top-decile bias 0.183) and the corrected posterior cuts it $\sim 7\times$ (0.026). The pre-registered null control — a one-parameter variance penalty — captures most of that decision-level benefit for a fraction of the machinery.
4. **The prediction–decision gap, reported (§5):** the corrected surrogate improves predictions (RMSE, tail bias) everywhere, yet does not beat plain last-value SH at matched budget — and on PD1 it is decisively *worse* at decisions despite better predictions, because refitting a prior on small pools adds decision noise the prediction metrics miss. We say so plainly; it is the calibrated null of §3 made concrete.

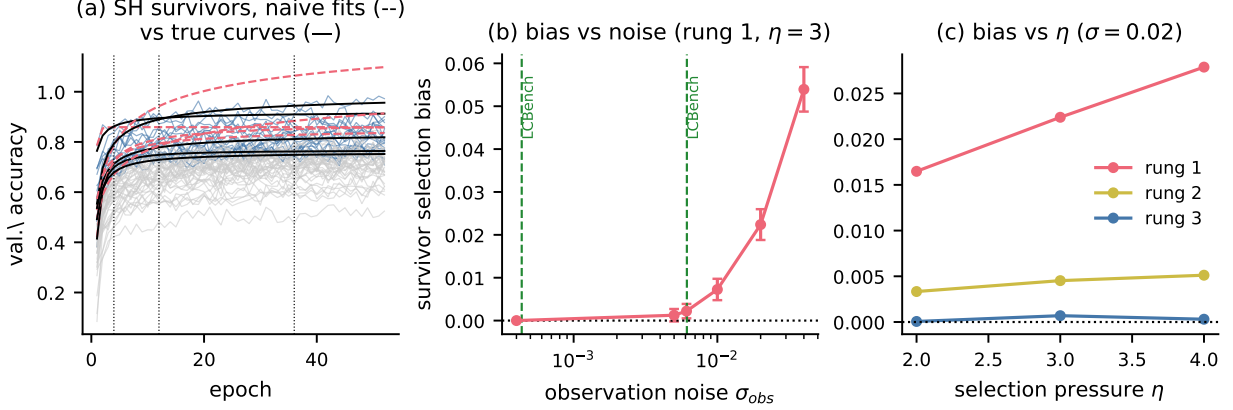


Figure 1: (a) The mechanism: SH survivors (blue) at the first rung were partly selected on upward noise; naive pow3 fits to their 4-epoch prefixes (dashed) overshoot their true continuations (black). (b) Survivor selection bias (survivor minus no-selection control) of naive extrapolation at the first rung vs observation noise, $\eta = 3$, 8 seeds; green dashed lines: noise levels calibrated from real LCBench curves (lag-1-difference median and pow3-residual median). (c) The bias grows with selection pressure η and decays across rungs as prefixes pin the fits down.

2 Setup

Curve family. Latent pow3 curves $y_i(t) = a_i - b_i t^{-c_i}$, $t = 1..T=52$; $\phi_i = (a_i, \log b_i, \log c_i) \sim N(\mu_{pop}, \Sigma_{pop})$ calibrated loosely to LCBench validation-accuracy curves; observations $\hat{y}_{it} = y_i(t) + \sigma \epsilon_{it}$, iid Gaussian. SH with rungs at $t \in \{4, 12, 36\}$, keep-fraction $1/\eta$ by the observed value at the rung epoch.

Estimators. *Naive:* per-curve nonlinear least squares (the standard extrapolating surrogate; 3, 10). *Tobit:* per-curve truncation correction — the survival events’ log Φ terms added to the per-curve NLL (no population prior; 19). *EB:* hierarchical Gaussian prior over ϕ fitted by marginal likelihood in a Laplace formulation (each curve summarized by its LS fit $\hat{\phi}_i$ with observation covariance $V_i = \sigma^2(J^\top J)^{-1}$, so $\hat{\phi}_i \sim N(\mu, \Sigma + V_i)$ in closed form); predictions are exact-likelihood posterior means via importance sampling with the Laplace posterior as proposal (the Laplace summary is only a proposal; its error cancels in the weights). *EB + Heckman:* the same, fitted on survivor curves only, with the selection correction $-n \log P_\psi(\text{survive all rungs})$ — the joint selection+outcome likelihood; survival probabilities are probit products over rungs, estimated by common-random-number Monte Carlo over the prior. A first attempt that Monte-Carlo-estimated the *data* marginal from prior samples collapses for peaked likelihoods (effective sample size ≈ 3 at $T = 52$, $\sigma = 0.01$); we document this failure mode and use the closed-form marginal instead.

3 B-E0: the gate — bias exists, is dose-responsive, and vanishes at clean-signal noise

For each rung we fit the naive surrogate on same-length prefixes of *all* configs (the no-selection control) and of survivors; the *selection bias* is the difference of their mean prediction errors, isolating censoring from generic extrapolation error. Results (Fig. 1; $n = 1000$ configs, 8 seeds): at the first rung ($t = 4$, where SH kills two thirds of the pool and the surrogate extrapolates 13 $\times$) the selection bias is 0.0012 ± 0.0015 at $\sigma = 0.005$, 0.0073 at 0.01 , 0.0224 at 0.02 , and 0.0539 at 0.04 accuracy units — monotone in σ , as the mechanism demands — and monotone in selection pressure ($0.0165 \rightarrow 0.0224 \rightarrow 0.0279$ at $\sigma = 0.02$ for $\eta = 2, 3, 4$). By the third rung the bias is an order of magnitude smaller ($0.0224 \rightarrow 0.0007$ at $\sigma = 0.02$): long prefixes leave little room for noise-driven selection to distort the fit. The pre-registered gate — bias exists and grows with noise and selection pressure — **passes**.

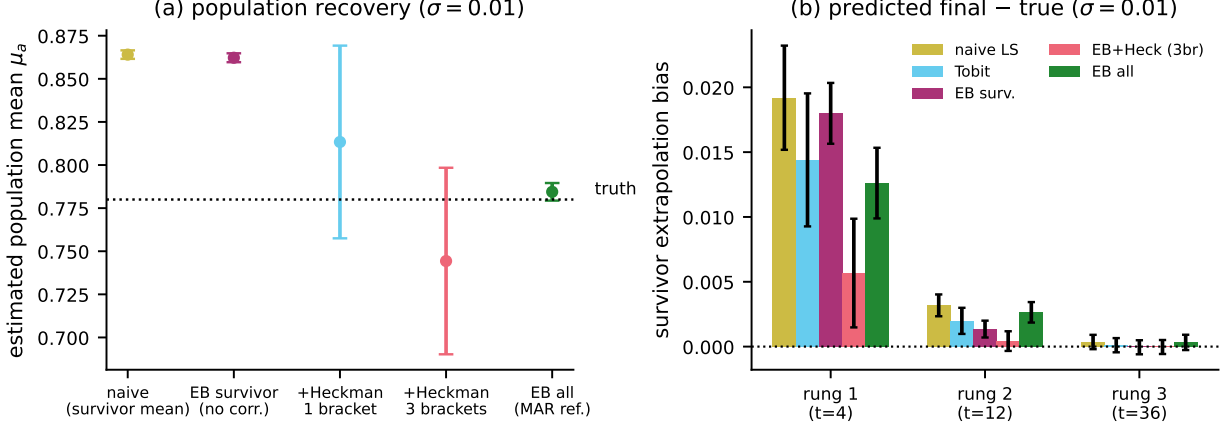


Figure 2: $\sigma = 0.01$, 8 seeds, mean $\pm$ sd. (a) Recovering the unselected population mean μ_a from survivor-only data: the naive survivor mean and uncorrected EB prior are biased upward; the survival-corrected fit from ONE bracket is weakly identified (high variance); THREE Hyperband-style brackets (rung-assignment randomness as exclusion restriction) roughly halve the bias but overshoot downward; the all-prefixes EB reference (using killed configs’ prefixes) is nearly unbiased and is the estimator we recommend. (b) Survivor extrapolation bias by rung.

The informative null. Calibrating σ from real LCBench curves gives a median lag-1-difference noise of 0.0004 (much of LCBench’s measurement error is a *persistent* per-config validation-set effect rather than per-epoch jitter) and a pow3-residual scale of 0.0061; at these levels the first-rung selection bias is 0.0000 and 0.0022 respectively — negligible to small. The honest conclusion, pre-registered as an acceptable outcome: *at LCBench-median noise, SH’s censoring is close to ignorable, and the practical bite of survivor bias is confined to noisy regimes* — small validation sets, quantized metrics, or tasks like PD1’s ImageNet ($\sigma \approx 0.02$ by the same calibration), which sits squarely in the dose-responsive zone.

4 B-E1: the corrected surrogate and its identification

Population recovery. Fitting the extrapolator’s prior on completed (survivor) runs is common practice; it estimates the *survivor* population. At $\sigma = 0.01$, three brackets of 999 configs leave 481 completed curves whose naive mean asymptote is 0.864 against a true 0.780 (uncorrected EB prior: 0.862), a +0.084 upward bias. Adding the survival-likelihood correction needs cross-bracket variation to bite at all: one bracket gives 0.813 ± 0.056 (weakly identified — the conditional likelihood *prefers* the biased solution, which we verified by evaluating it at truth vs. at the naive fit and finding the naive fit scores lower). Three brackets give 0.744 ± 0.054 , cutting |bias| from 0.084 to 0.036 — but now overshooting *below* truth, and increasingly so with noise (0.76, 0.74, 0.66 at $\sigma = 0.005, 0.01, 0.02$): the functional-form correction is itself sensitive to the assumed noise model, the classic Heckman fragility [16]. The population sd is recovered well (0.076 vs true 0.070). The estimator we actually recommend is the MAR-correct one that refits the prior on *all* observed prefixes, killed configs included: it lands at 0.784, essentially unbiased, and it is exactly the data a replaying tool already holds. The parallel with the companion paper’s instrument-absent honesty curve [8] is exact — same machinery, same fragility — but the practical lesson here is sharper: because selection is on observed prefixes, one need not model it at all; one need only fit on the prefixes rather than on the survivors.

Survivor extrapolation. At the first rung the naive fit’s bias is 0.0192 ± 0.0040 ; the per-curve Tobit correction (no prior) shaves it to 0.0144 — the truncation terms only touch the marginal survivors — while the exact posterior under the corrected prior essentially removes it (0.0057; RMSE 0.0374 vs 0.0700 naive). Since survival is a deterministic function of the observed prefix, the *posterior* needs no extra selection term — exact Bayes has zero conditional bias given any data-measurable selection event; what selection breaks is the *prior* estimated from survivors, and that is where the correction belongs. This decomposition is, to our knowledge, not stated in the LC-surrogate literature.

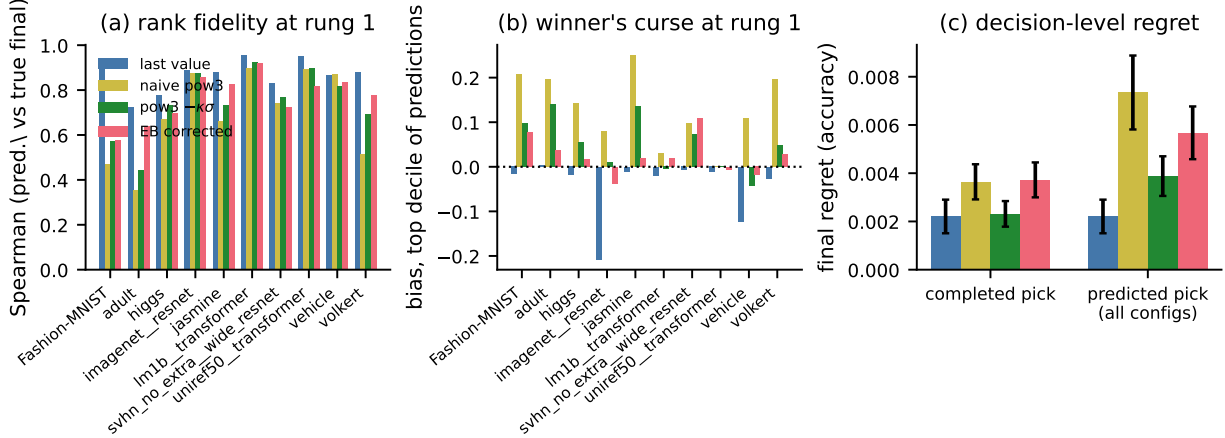


Figure 3: Replays on real curves (six LCBench datasets, four PD1 tasks; 8 seeds $\times$ pool sizes 50/200). (a) Rank fidelity of final-value predictions from first-rung prefixes. (b) The winner’s curse: mean prediction error in the top decile of each surrogate’s own predictions. (c) Decision-level regret under SH replay with surrogate-driven promotion, and under picking the best *predicted* config from all prefixes including killed ones.

Arellano–Bond variant. The lag-instrument dynamic-panel idea [2] — extrapolating increments with second-lag instruments — gives first-rung bias 0.1125, worse than naive least squares (power-law curvature defeats short-lag AR structure). Per the pre-registered plan we drop it without ceremony.

Fresh-config transfer. Priors learned from survivor-only data and used on *new* unselected configs (the “surrogate trained on completed runs” deployment) show the same ordering but a small effect at the 12-epoch prefix we test, where even the naive fit extrapolates well: bias 0.0054 uncorrected vs -0.0006 corrected (0.0013 for the MAR reference). The transfer penalty of a survivor-fit prior is real but modest once prefixes are moderately long.

5 B-E2–E4: LCBench and PD1 replays

We replay synchronous SH (rungs at fixed fractions of each task’s horizon, $\eta = 3$) on precomputed curves — six LCBench datasets [22] and four PD1 tasks [20] — with the surrogate driving both promotions and the final pick. Surrogates: last-value (standard SH), naive pow3, pow3 with a one-parameter variance penalty $\hat{y} - \kappa \text{sd}$ (the pre-registered B-E4 null control, $\kappa = 1$), and the EB-corrected surrogate fitted on the fly from all prefixes observed so far in the replay. FT-PFN [17] is cite-compared rather than run: its synthetic-prior pretraining mitigates per-curve overfitting — a shrinkage effect our EB model makes explicit — but does not model the deployed sampler’s selection process, which is the object of study here.

Predictions. From first-rung prefixes, last-value ranks configs surprisingly well (Spearman 0.84 on LCBench) but is uniformly pessimistic (curves still rise); naive pow3 collapses the ranking (0.59) and exhibits a large winner’s curse — the top decile of its own predictions over-promises by 0.183 accuracy units — while the corrected surrogate recovers much of the rank fidelity (0.73) with the best RMSE (0.095 vs 0.102 naive) and a $\sim 7\times$ smaller tail bias (0.026 vs 0.183). PD1 mirrors the prediction pattern (Spearman 0.91 / 0.85 / 0.83; top-decile bias 0.053 naive vs 0.021 corrected).

Decisions. Here the story splits by benchmark, and we report the split. Under promotion-by-surrogate on LCBench, last-value SH is already near-optimal (regret 0.0020), consistent with the §3 null at LCBench noise: naive pow3 promotion is *worse* (0.0057; paired Wilcoxon vs corrected $p = 0.0001$), and the corrected surrogate pulls extrapolation-driven promotion back to last-value parity (0.0024; vs last-value $p = 0.2310$, not significant). On PD1, however, the corrected surrogate improves *prediction* yet *worsens decisions*: its

promotion regret (0.0078) is significantly *above* last-value’s (0.0017; $p = 0.0000$) and above naive pow3’s (0.0050) — refitting a prior on small, heterogeneous PD1 pools each rung adds decision noise the prediction metrics do not see. This is the prediction–decision gap in its starkest form, and it is exactly what the pre-registered kill condition told us to watch for. When the final pick may choose *any* config from its predicted final — where killed configs’ extrapolations compete — naive’s winner’s curse costs real accuracy (regret 0.0089; it picks an early-killed config in 26% of LCBench replays), and both the cheap variance penalty (0.0046) and the corrected surrogate (0.0035) roughly halve that damage — but neither reaches plain last-value’s 0.0020. That is the B-E4 verdict, stated plainly: *the one-parameter variance penalty captures most of the correction’s decision-level benefit, and neither intervention beats plain last-value SH at matched budget on these benchmarks*. The corrected surrogate’s value on these families is confined to (i) whenever extrapolating surrogates are used at all — it makes them safer — and (ii) unbiased final-value prediction itself (reporting, meta-learning, transfer), not raw SH regret at clean-signal noise.

6 What we do not claim

We claim nothing beyond the tested benchmark families (LCBench tabular MLP curves; four PD1 tasks) and the pow3 curve family; other curve shapes (double-descent, cyclic schedules) violate the parametric prior. We do not claim corrected surrogates improve end-to-end HPO regret at LCBench-median noise — they do not, and §3 explains why that is the expected outcome, not a failure of the theory. We do not claim asynchronous ASHA results; replays are synchronous SH. The rank-based rung thresholds are treated as fixed constants in the survival likelihood (standard, but an approximation). We make no claim about FT-PFN’s internals beyond the cited design.

7 Limitations

Observation noise is modeled iid Gaussian per epoch; real LCBench noise has a persistent per-config component (fixed validation set) that acts like latent quality for within-run decisions and is invisible to our lag-1 calibration — our two calibration estimates bracket it. The Laplace summary degrades for 3–4-point prefixes (we correct predictions with exact-likelihood importance sampling, but the prior fit inherits some coarseness). The multi-bracket estimator assumes bracket assignment is random — true in Hyperband by construction, but not in adaptive schedulers. Survivor counts below ~ 50 leave the survival-corrected fit weakly identified even with brackets.

8 Reproducibility

All code, tests (including the SH harness, Tobit direction, and multi-bracket recovery tests), committed result JSONs, and the auto-generated `numbers.tex` (regenerate-and-diff passes at submission) are in the repository; discipline inherited from Howe [7]. LCBench data [22] and PD1 [20] are public; converter scripts and calibration JSONs are committed. Total compute: under three GPU-hours plus CPU replays on a 16-core desktop. Archived release: DOI 10.5281/zenodo.TBD (filled at submission). Pre-submission recency sweep logged in PLAN.md.

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